

Alcohol-related mortality in 15–24-year-olds in Denmark (2010–2019): a nationwide exploratory study of circumstances and socioeconomic predictors



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Summary

Background Adolescents and young adults aged 15–24 years are disproportionately affected by unnatural deaths, including accidents, suicide and interpersonal violence for which alcohol is a leading risk factor. We aimed to explore the extent of and circumstances surrounding alcohol-related deaths in young people aged 15–24 years and whether socioeconomic background and ethnicity differ in those who died due to alcohol-related causes as compared to the background population.

Methods All deaths of 15–24-year-olds occurring in Denmark from 2010 to 2019 were investigated. We manually reviewed death certificates containing information on circumstances, results from autopsies and blood tests, and statements from witnesses. Relevant information published in the media (most often newspaper articles) was included. Our main outcome measures were alcohol-related death and manner of death (accidents (transport accidents, drownings, falls, poisonings), suicide and violence). Further, we designed a population-based case-control study including 10 age- and sex-matched controls per case to test whether there was a socioeconomic gradient in alcohol-related deaths. We used parents' educational level and employment status to define socioeconomic position. Immigration status was used to assess ethnicity.

Findings Over the 10-year period, 1783 deaths occurred among 15–24-year-olds. Of those, 1067 (60%) were due to unnatural causes, corresponding to a mortality rate of 14.8 (95% confidence interval: 13.9–15.7) per 100,000. Twelve percent of unnatural deaths ($n = 125$) were alcohol related, corresponding to a rate of 1.7 (1.4–2.0) per 100,000, and were higher in males (2.9 [2.3–3.4]) than in females (0.6 [0.3–0.8]); thus, males accounted for 105 (84%) of alcohol-related deaths. The majority of alcohol-related deaths occurred on Fridays, Saturdays and Sundays ($n = 77$, 62%). Accidents accounted for 82% ($n = 102$) of alcohol-related deaths, followed by suicide ($n = 19$, 15%) and interpersonal violence ($n = 4$, 3%). Of all fatal accidents, 102 of 636 (16%) were alcohol related. Of all deaths caused by drownings and falls, 14 of 26 (54%) and 10 of 25 (40%), respectively, were alcohol related. Alcohol-related drownings most often occurred while the deceased was alone, whereas alcohol-related falls most often occurred in relation to parties, involving falls from a window or balcony. Those who died from alcohol-related causes more often had parents with a short education or who were unemployed, as compared to the general population. For example, odds ratios were 3.9 (2.2–7.0) and 1.8 (1.2–2.9) for having parents with short and medium as compared to long educations. The odds ratio for being of Danish origin was 4.0 (1.7–9.5) compared to being first- or second-generation immigrants.

Interpretation In 15–24-year-olds, alcohol-related deaths accounted for a substantial proportion of all unnatural deaths. There was substantial socioeconomic inequality in alcohol-related deaths, as has repeatedly been shown for chronic alcohol-related mortality in older adults.

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Research in context

Evidence before this study

We searched PubMed with the terms (adolescent or young adults or youth) and (death or mortality) and (alcohol), for papers published between Jan 1, 1970, and Aug 30, 2022. No language restrictions were applied.

Studies on alcohol-related deaths in adolescents and young adults were scarce. Though it is known that alcohol is a causal factor in many circumstances, studies focused on alcohol in relation to one particular cause of death. One study identified all teenage deaths occurring between 1981 and 2000 in a northern region of Sweden and found that 28% tested positive for alcohol. However, the causal role of alcohol was not investigated further.

Added value of this study

By combining multiple data sources such as death certificates, information published in common media and nationwide register information we investigated all deaths of 15–24-year-olds occurring in Denmark from 2010 to 2019 and found that

twelve percent of unnatural deaths were causally related to alcohol. Most alcohol-related deaths happened among males and accidents were the most common manner of death. Additionally, we found a clear social gradient and alcohol-related deaths were much more frequent among adolescents and young adults with lower socio-economic position, than adolescents and young adults with higher socio-economic position and less frequent among those who were first- and second-generation immigrants as compared to those of Danish origin.

Implications of all the available evidence

This study provides a comprehensive description of circumstances surrounding alcohol-related deaths among adolescents and young adults. Our findings indicate that there continues to be an urgent need for public health interventions aimed at reducing alcohol intake among adolescents and young adults.

Introduction

Adolescents and young adults aged 15–24 years appear to have a slower decline in mortality compared to young children.¹ This age group is disproportionately affected by unnatural deaths, including accidents, suicide and interpersonal violence.^{1–3} Alcohol is a leading global risk factor for such events.^{4,5}

Alcohol may cause harm in several ways. Alcohol intake, especially binge drinking, leads to impaired motor skills and poor balance together with a lower risk aversion. This in turn increases the risk of acute consequences, such as actions that are regretted afterwards (e.g., conflicts and unprotected sex) or accidents which in worst case scenario are fatal.^{6,7}

Studies on alcohol-related deaths in adolescents and young adults are few. Johansson et al. examined unnatural deaths occurring in teenagers in a northern region of Sweden between 1981 and 2000. They identified 355 unnatural deaths, and among these 28% tested positive for ethanol. However, the course of action leading to the deaths was not uncovered, which ultimately is required to assess whether a death is related to alcohol or not. Other studies have primarily focused on a single cause of death, such as drownings or road traffic injuries,^{8–16} and then elucidated reasons for that specific cause.

In mortality statistics, some under-reporting of alcohol as cause of death has been documented.¹⁷ Such lack of completeness is likely to be more frequent when alcohol is the contributing cause, for instance, in cases of drownings and suicides, as opposed to being the underlying cause, such as in cases of alcohol poisonings where death is directly caused by the intoxication. In cases where alcohol is a contributing cause, information

on the course of action and circumstances surrounding the death may be needed to make a final decision about whether or not the death was attributable to alcohol.

In adults, it has been repeatedly shown that alcohol-related mortality is socially skewed, being three to five times higher in lower socioeconomic groups compared to those with a high socioeconomic position.¹⁸ Typically, mortality from chronic alcohol-attributable diseases such as liver cirrhosis or alcoholic cardiomyopathy has been studied. These conditions are pre-empted by years of heavy drinking, thus are rare in adolescents and young adults, but it is unknown whether similar patterns exist in alcohol-related deaths among adolescents and young adults.

Therefore, further research is needed to assess alcohol-related deaths specifically among adolescents and young adults. It is necessary to combine data from different sources to gain a comprehensive understanding of alcohol-related deaths among adolescents and young adults.

In this study, we aimed to assess the extent of and circumstances surrounding alcohol-related deaths among adolescents and young adults aged 15–24 years. Furthermore, we aimed to study whether socioeconomic background differs in alcohol-related deaths in adolescents and young adults compared to the background population.

Methods

We conducted two substudies: First, we study all deaths occurring in 15–24 year olds from 2010 to 2019 olds to decide the number of alcohol-related deaths and the circumstances surrounding these deaths. Second, we

undertake a population-based matched case-control study, matching each alcohol-related death to controls, to compare the socioeconomic profile in alcohol-related deaths with the socioeconomic profile of the general population. Data sources were The Danish Register of Causes of Death¹⁹ and individual death certificates, all administered by The Danish Health Data Authority. Furthermore, we searched for supplementary information from media coverage. We also used The Danish Civil Registration System,²⁰ The Population's Education Register,²¹ The Labour Force Statistics,²² and immigration data from Statistics Denmark.²³

Mortality data

For all deaths occurring in Denmark, a death certificate is filled out. The certificate is based on a post-mortem examination performed by a medical doctor.¹⁹ The purpose of the post-mortem examination is to ensure that death has occurred with certainty, establish facts about the death and determine whether the cause of death was natural, that is caused by a medical condition or disease, or unnatural, that is, due to an external factor such as an accident, suicide or a criminal offence.

In the death certificate, the underlying cause of death, and in some instances also contributing causes, is stated.¹⁹ The underlying cause of death is the disease or injury that started the chain of events ultimately leading to death, whereas a contributing cause is a condition contributing to the fatal outcome but not directly causing the death.²⁴ For example, if a person drowns while intoxicated, the underlying cause is drowning whereas the intoxicated state is a contributing cause.

In-depth post-mortem examinations are performed in cases of unnatural death if a person is found dead, if death occurred suddenly and unexpectedly or if the death is of special interest for other reasons. In such cases, the physician contacts the police, who decide whether a forensic post-mortem examination should be performed. A forensic post-mortem examination is performed by a physician from the Danish Agency for Patient Safety under the surveillance of the police. At the forensic post-mortem examination, relevant information (police investigation, interrogations, medical records, etc.) is compared with an examination of the body to determine the manner and cause of death. In some cases, an autopsy is performed with the purpose of determining the exact cause of death and to gain knowledge of any condition, disease or diseases that the deceased might have suffered from.

In all cases, death certificates are filled in, with the systematic recording of key information, such as sex, age, date, place and manner of death, underlying and contributing causes and whether an autopsy has been carried out. This information is transferred to the Register of Causes of Death in which all deaths occurring in Denmark are registered.¹⁹ The Danish Health Data

Authority manually reviews information on all unnatural deaths to validate the registration and to verify that the underlying cause of death is stated correctly and in accordance with WHO's tenth International Classification of Diseases (ICD-10).

In addition, certificates may contain information obtained from forensic examinations and autopsies, toxicological analyses, statements from witnesses, friends and family, as well as extracts from the police report and medical charts. This information is all written in prose and not transferred to the Register of Causes of Death. Therefore, it can only be extracted by manually reviewing the certificate itself.

Review of death certificates

We gained access to death certificates of all deaths among 15–24-year-olds occurring in Denmark from 1 January 2010 to 31 December 2019 ($n = 1783$). We excluded cases wherein the manner of death was natural ($n = 716$) (Fig. 1), leaving 1067 certificates for review. From death certificates, we systematically extracted data on identification, sex, underlying cause of death, manner of death, place and time of death and time the deceased was found. If available, we extracted information about autopsy findings, mental illness, and substance abuse as well as other relevant information. Information about alcohol intake was of particular interest: We recorded alcohol concentration in body fluids (available in 76 cases) and whether alcohol or other substances were mentioned anywhere in the available material (death certificate, autopsy report, interviews with relatives, witness reports, etc.).

We included cases for further investigation if the underlying or contributing cause of death was alcohol related ($n = 80$), alcohol was mentioned in some shape or form in the death certificate ($n = 146$) or when a measurement of alcohol concentration in blood, urine or eye was positive ($n = 75$), in total 178 cases. Focusing further on information about alcohol and circumstances surrounding the death, we aimed at dividing the cases into three categories: alcohol related, potentially alcohol related and not alcohol related, depending on whether or not it was likely that alcohol was an underlying or contributing cause. If the deceased had been drinking and taken drugs prior to the death but there was doubt about whether the alcohol or drugs had been a crucial contributing cause of death, the death was categorised as potentially alcohol related. This was also the case for suicide victims in cases where it was difficult to determine whether alcohol intake was the deciding factor. Deaths categorised as potentially alcohol-related were not included for further analysis.

Alcohol-related deaths were:

- Deaths that already had an alcohol-related cause of death stated in the Register of Causes of Death associated either as underlying or contributing cause

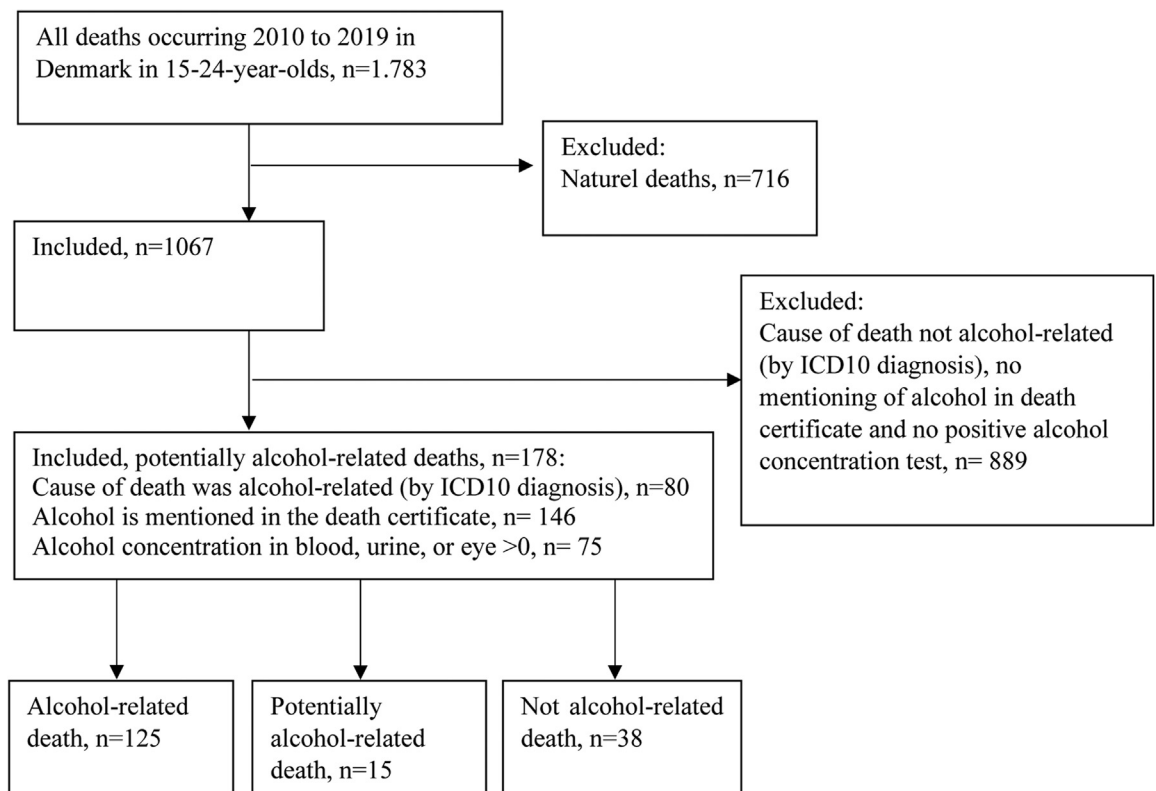


Fig. 1: Flowchart for the identification of alcohol-related deaths occurring 2010 to 2019 in Denmark in 15-24-year-olds, by manual review of all death certificates.

of death. We found no evidence to contradict the diagnosis in any of these cases. A full list of ICD10-codes can be found in [Supplementary Table S1](#).

- Deaths resulting from an accident wherein the deceased had a clear indication of intoxication or a BAC >0.05, as it was assumed that above this limit motor skills as well as the willingness to take risks was so sufficiently affected that alcohol was considered a part of the causal mechanism. It was also a criterion that the deceased was an active part of the accident, meaning that alcohol-intoxicated passengers in car accidents were not counted as alcohol-related deaths.
- Suicide, when alcohol consumption either alone or mixed with medication was considered to have led to a change for the worse in state of mind, ultimately triggering the fatal action.
- Acts of violence or homicide during which the deceased had been so intoxicated that it was assessed that they were unable to react to the situation.

SH and UB independently reviewed all the information and determined whether or not the cause of death was alcohol related (yes, no, possibly). After this, they compared all cases and reached consensus on

whether the cause of death was alcohol related for each death.

Information from media coverage

We enriched the data from the identified cases if alcohol-related deaths with publicly available information from media and press releases. Information retrieved from death certificates on age, date, place and cause of death and so on was used to identify cases that had been described publicly, using freely available search engines. The extra information gathered this way included details concerning place of death and circumstances surrounding the incident, such as whether the deceased was alone or with friends.

Population-based case-control design

To test whether there was a socioeconomic gradient in alcohol-related deaths, we used a population-based matched case-control design. All Danes are given a personal identifying number at birth or immigration, enabling linkage at individual level between national administrative, clinical, health and social registries.²⁰ Information on all Danes is included in the Danish Civil Registration System (DRS), which contains information on sex, date of birth and death, immigration,

and nationality. The personal identification number also permits the identification of parents.

Using the Danish Civil Registration System, ten controls were drawn from the entire Danish population with the same sex and age (± 50 days) at the same time of death as the alcohol-attributable cases.

Socioeconomic position and ethnicity

We used parents' educational level (Population's Education Register)²¹ and employment status (Register-based Labour Force Statistics)²² to define socioeconomic position. The highest education attained by either of the parents was categorised into three groups as short (elementary school only), medium (high school and vocational education and short tertiary education) and long (medium and long tertiary education corresponding to university bachelor and master), and employment status was categorised into three groups (both parents working; one parent working, one receiving benefits; and both parents receiving benefits). Information on ethnicity was obtained from Statistics Denmark,²³ and ethnicity was defined as Danish origin or first- or second-generation immigrant.

Statistical analysis

Stata version 17 was used for analysis. Mortality rates were calculated using publicly available information about sex and age specific population sizes in the third quarter of each year for the period 2010–2019. We reported 95% confidence intervals [CI] for all rates.

Logistic regression was used to compare indicators of socioeconomic position (parental education and employment) and ethnicity in alcohol-related deaths and controls. All analyses included matching parameters (sex and age), and clustered standard errors were specified, allowing for intragroup correlation within matching clusters to relax the assumption that individuals were independent. Tests for statistical significance in the distribution of deaths by weekday versus weekend and by season were done by a Chi-square test. To test for trend over time in number of alcohol-related deaths, the correlation between yearly number of events and year was assessed using the Spearman correlation coefficient.

Role of the funding source

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Results

Between 1 January 2010 and 31 December 2019, there were 1783 deaths among 15–24-year-olds in Denmark. Of these, 716 (40%) were natural and 1067 (60%) were unnatural. The overall mortality rate was 24.8 [95% CI

23.6 to 25.9] per 100,000. The rate was higher among 20–24-year-olds (29.7 [95% CI 27.9 to 31.4]) than among 15–19-year-olds (19.7 [95% CI 18.2 to 21.2]) (Table 1).

In the manual review of the 1067 death certificates from unnatural deaths, we found that alcohol was an underlying or contributing cause in 125 fatalities (Table 2). Drugs or use of medication was mentioned in 33 of the 125 deaths. Of these, 79 deaths (64%) were also registered in the Register of Causes of Death with alcohol stated as the underlying or contributing cause of death, indicating that there was an under-reporting of alcohol-related deaths in the register. There was no apparent systematic difference between deaths that were registered as being alcohol-related and those that were not. In 15 of the deaths not registered as being alcohol-related, alcohol tests were performed, and all were positive (median = 1.45 o/oo, range 0.57–2.81).

Alcohol-related deaths accounted for 12% of all unnatural deaths in 15–24-year-olds (Table 2). Mortality rates for alcohol-related death were 2.9 per 100,000 in males and 0.6 per 100,000 in females (Fig. 2). The mean annual number of alcohol-related deaths was 12.5, averaging one alcohol-related death per month during the 10-year period (Supplementary Fig. S1). There was a downward trend in the number of yearly events as indicated by a Spearman correlation coefficient of -0.60 , however not statistically significant ($p = 0.07$). Alcohol-related deaths happened most often during the weekend: 62% happened on a Friday, Saturday, or Sunday. Had the alcohol-related deaths been equally distributed across all days of the week, 43% would have happened on those three days ($p < 0.0001$) (Fig. 3). We also investigated whether there was a seasonal variation in alcohol-related deaths; however, no clear seasonal pattern was observed ($p = 0.79$) (Supplementary Fig. S2).

Approximately three out of four unnatural deaths happened among males, and accidents were the most common manner of unnatural deaths, accounting for 58% of unnatural deaths not related to alcohol and 82% of alcohol-related deaths, followed by suicide and interpersonal violence (Table 2). For all types of manner of death, rates were higher in males than in females (Fig. 4).

Transport injuries (as pedestrian or involving bicycle, motorcycle, moped or car) were the most common cause of death by accidents in 15–24-year-olds, and alcohol accounted for 12% of those (Fig. 5). For drownings and falls, 54% and 40%, respectively, could be attributed to alcohol.

We further investigated circumstances surrounding alcohol-related drownings. Seven of the alcohol-related drownings happened at a harbour, whereas the remaining seven happened in other locations. Combining information from witness statements and a mass media search, we found that 11 of the 14 alcohol-related drownings happened immediately after the

	Total deaths		Natural deaths		Unnatural deaths	
	N (%)	Rate ^a (95% CI)	N (%)	Rate ^a (95% CI)	N (%)	Rate ^a (95% CI)
Total	1783	24.8 (23.6–25.9)	716	10.0 (9.2–10.7)	1067	14.8 (13.9–15.7)
Age group						
15–19 years	691 (39)	19.7 (18.2–21.2)	289 (40)	8.2 (7.3–9.2)	402 (38)	11.5 (10.3–12.6)
20–24 years	1092 (61)	29.7 (27.9–31.4)	427 (60)	11.6 (10.5–12.7)	665 (62)	18.1 (16.7–19.4)
Sex						
Male	1233 (69)	33.5 (31.6–35.4)	413 (58)	11.2 (10.1–12.3)	820 (77)	22.3 (20.8–23.8)
Female	550 (31)	15.7 (14.4–17.0)	303 (42)	8.6 (7.7–9.6)	247 (23)	7.0 (6.2–7.9)

^aIncidence per 100,000.

Table 1: Number and rates of total, natural, and unnatural deaths occurring 2010 to 2019 in Denmark in 15–24-year-olds, by age and sex.

	Unnatural deaths			
	Not alcohol-related		Alcohol-related	
	N (%)	Rate ^a (95% CI)	N (%)	Rate ^a (95% CI)
Total	927 (88)	12.9 (12.1–13.7)	125 (12)	1.7 (1.4–2.0)
Age group				
15–19 years	357 (39)	10.2 (9.1–11.2)	41 (33)	1.2 (0.8–1.5)
20–24 years	570 (61)	15.5 (14.2–16.8)	84 (67)	2.3 (1.8–2.8)
Sex				
Male	701 (76)	19.1 (17.6–20.5)	105 (84)	2.9 (2.3–3.4)
Female	226 (24)	6.4 (5.6–7.3)	20 (16)	0.6 (0.3–0.8)
Manner of death				
Accident	534 (58)	7.4 (6.8–8.1)	102 (82)	1.4 (1.1–1.7)
Suicide	331 (36)	4.6 (4.1–5.1)	19 (15)	0.3 (0.1–0.4)
Interpersonal violence	56 (6)	0.8 (0.6–1.0)	4 (3)	0.1 (0.0–0.1)

Note: 15 deaths were categorized as potentially alcohol-related and not included why the sum of unnatural deaths not related to alcohol and alcohol-related deaths does not sum up to total number of unnatural deaths. ^aIncidence per 100,000 persons.

Table 2: Number and rates of unnatural deaths not related to alcohol and alcohol-related deaths occurring 2010 to 2019 in Denmark in 15–24-year-olds, by age, sex, and manner of death.

deceased had attended a party or a night out with friends, and of those deaths, eight were last seen or gave their last sign of life between 11:00 p.m. Friday night and 06:00 a.m. Sunday morning. In 12 of the 14 drowning deaths, the deceased was alone when he or she drowned. In contrast to alcohol-related drownings, the majority of fatalities caused by falls occurred while the deceased was not alone (seven of ten fatalities). In those cases, the deceased fell from a balcony or window during a party or some gathering of friends.

Comparing the socioeconomic background of 15–24-year-olds who died from alcohol-related causes with population-based controls, we found a clear social gradient meaning that young people who died from alcohol-related causes more often had parents with low socioeconomic position as compared to population controls. Odds ratios were 3.9 (95% CI 2.2 to 7.0) and 1.8 (95% CI 1.2 to 2.9) for having parents with short and medium as compared to long education (Table 3). Likewise, young people who died from alcohol-related causes more often had parents who were receiving

unemployment benefits as compared to having parents who were working (Table 3). Young people who died from an alcohol-related cause were four times more likely to be of Danish as compared to non-Danish ethnicity (Table 3).

Discussion

After manual reviews of death certificates of all unnatural deaths among 15–24-year-olds in Denmark between 1 January 2010 and 31 December 2019, we found 125 alcohol-related deaths, accounting for 12% of all unnatural deaths. This means that in Denmark every month during this 10-year period a young person died from an alcohol-related death. Alcohol-related deaths were more common among males than females, and four out of five alcohol-related deaths were accidents. 16% of all fatal accidents were alcohol related. Of deaths caused by drownings and falls, 54% and 40%, respectively, were alcohol related. Alcohol-related drownings most often occurred while the deceased was alone after

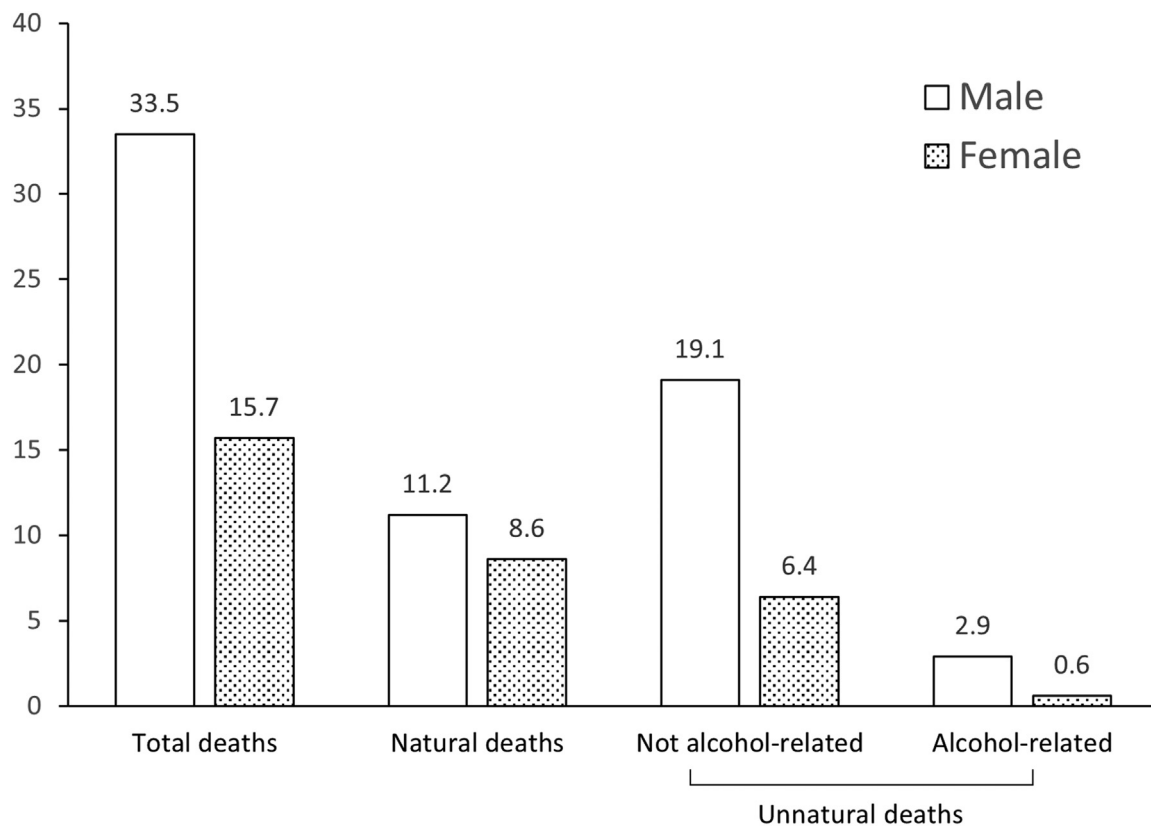


Fig. 2: Rates (per 100,000) of total, natural and unnatural deaths (not alcohol related and alcohol-related) occurring 2010 to 2019 in 15-24-year-olds in Denmark, by sex.

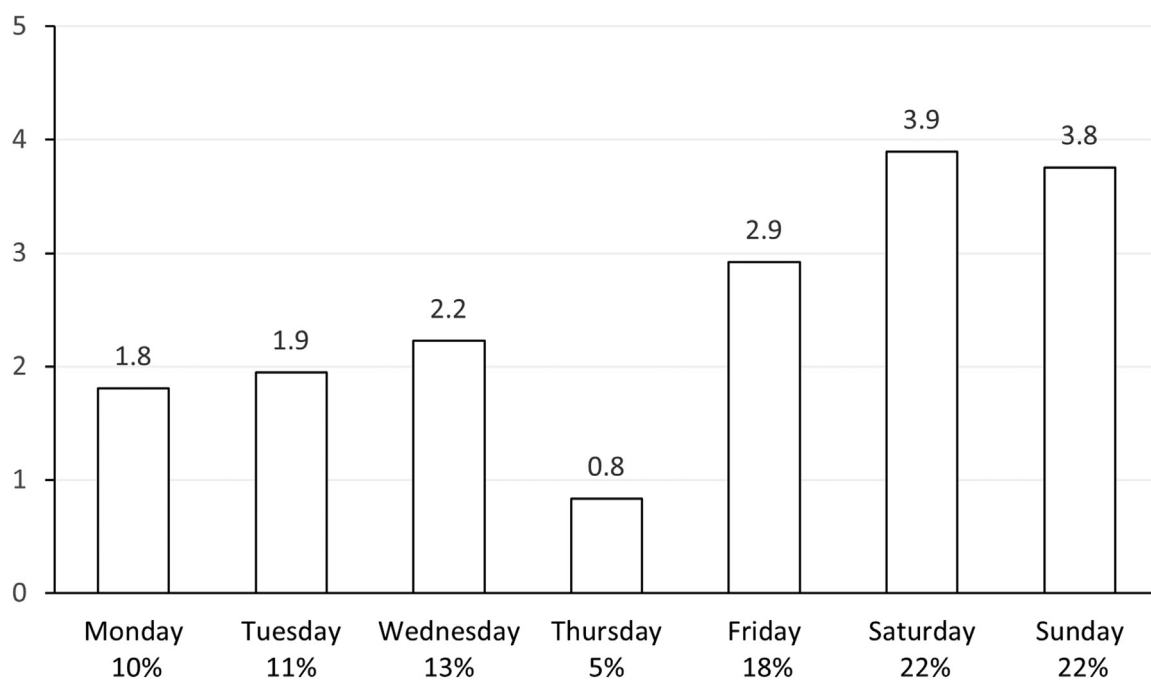


Fig. 3: Alcohol-related mortality rates (per 1,000,000) occurring 2010 to 2019 in 15-24-year-olds in Denmark, by day of week.

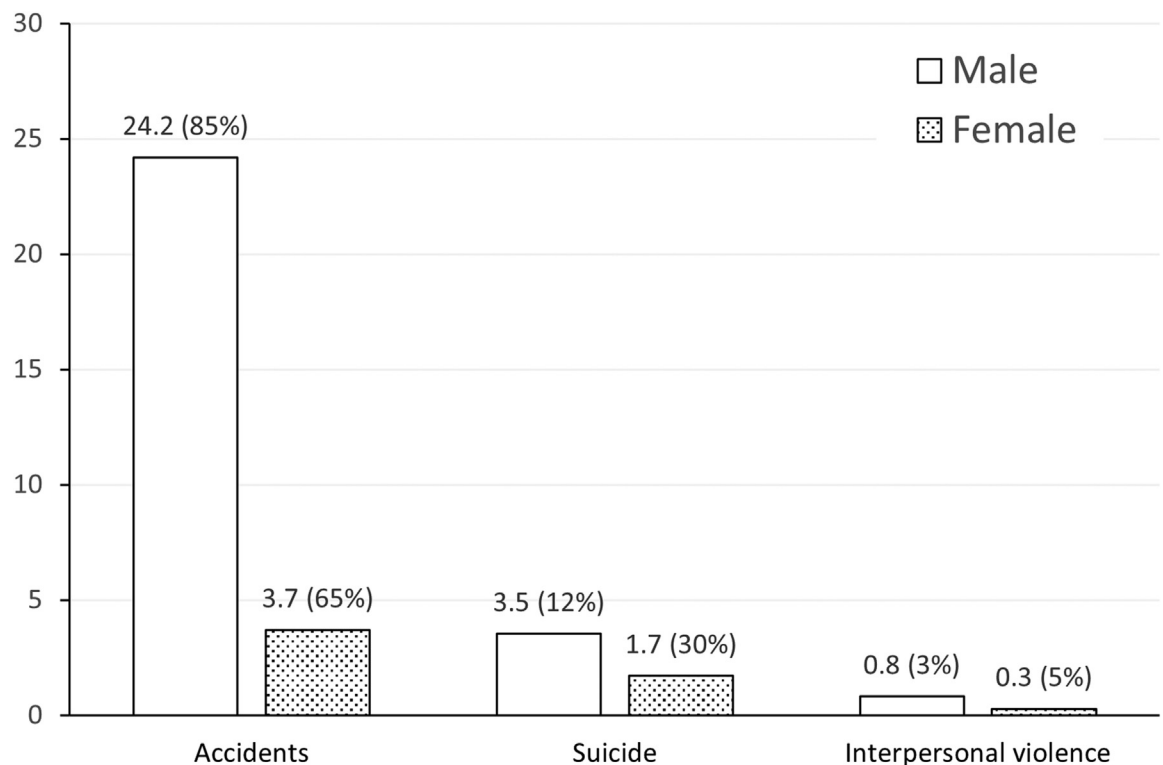


Fig. 4: Alcohol-related mortality rates (per 1,000,000) and percent within each sex, occurring 2010 to 2019 in Denmark in 15–24-year-olds, by manner of death and sex.

a party, whereas alcohol-related falls most often occurred in relation to parties. Furthermore, we found a high degree of social inequality, as those who died from alcohol-related causes were four times more likely to have had parents with short education and who were unemployed, as compared to the general population. Adolescents and young people who themselves or their parents were emigrants had a much lower risk of alcohol-related death compared to those of Danish origin.

Previous studies in Sweden and Ireland found higher proportions (28% and 52%, respectively) of alcohol-related deaths among all unnatural deaths than we did in this study.^{25–27} The large variation in the proportion of alcohol-related deaths is mainly due to differences in the definition of alcohol-related deaths and the study population as well as the systematics of data extraction. In our categorisation, we consider whether alcohol was an underlying or contributing cause of death, and we used 0.5 as our BAC threshold, which is also the legal alcohol limit for driving in most European countries as well as in Denmark. The three other studies did not consider whether alcohol was part of the causal mechanism, and they had a lower threshold for BAC. Using our definition, we can give an estimate of how many deaths could have potentially been prevented

if the alcohol intake were reduced, and therefore we find this definition more useful in terms of prevention of alcohol-related deaths.

Like previous studies, we found that the incidence of alcohol-related deaths was higher among males than females.^{25,27–29} The most common manner of alcohol-related deaths was accidents, among both males and females, but among females a higher proportion of alcohol-related deaths were suicides than among males. This finding is in line with a study that found that males more often presented with alcohol-related violence or other unspecified injuries, whereas females more often presented with alcohol-related self-harm or intoxication in emergency departments.³⁰

Our results show that social inequality in alcohol-related deaths exists among adolescents and young adults, which matches the pattern found among adults.³¹ Our results are in line with a Danish study that found that alcohol-related harm, such as getting into a fight or having sex and regretting it afterwards, happened more often among high school students with lower socioeconomic backgrounds, even when the level of alcohol consumption was accounted for.⁷ This indicates that the negative consequences of drinking alcohol are more pronounced in adolescents and young adults with a lower socioeconomic position. More alcohol-related

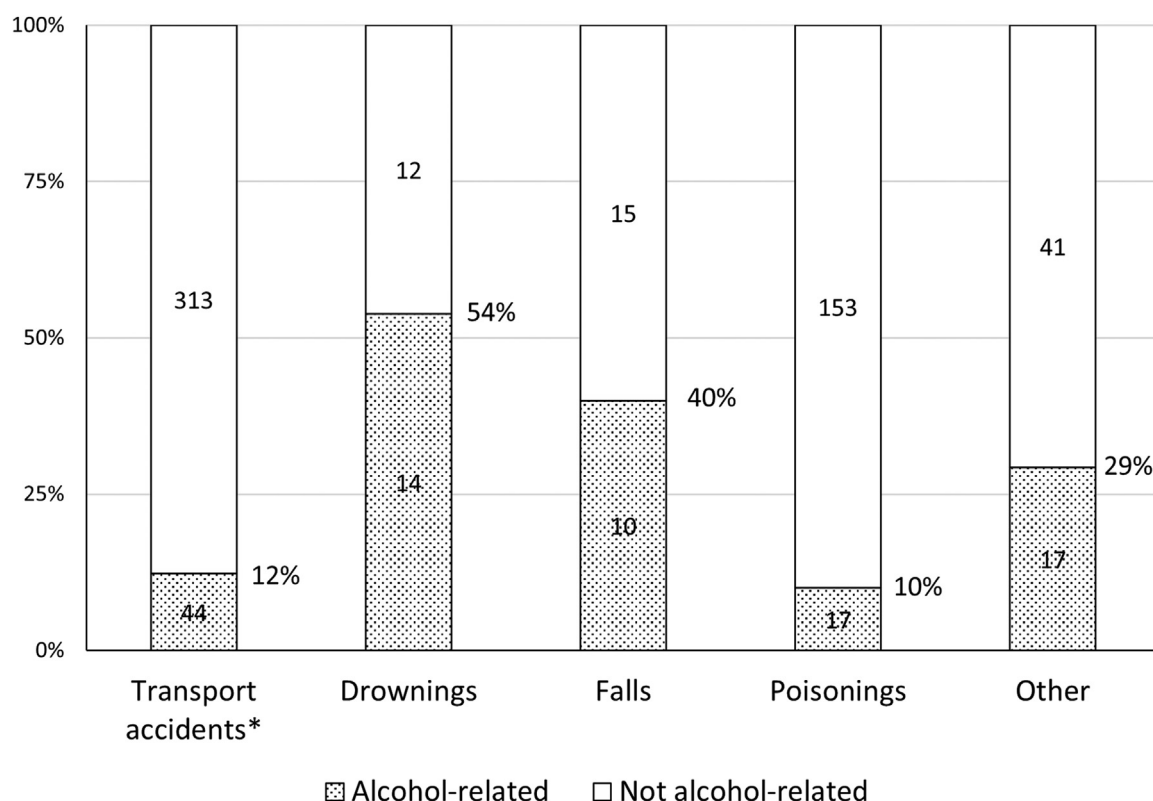


Fig. 5: Type of accident in 636 alcohol-related and not alcohol-related fatalities occurring 2010–2019 in 15–24-year-olds in Denmark.

*Transport accidents include car, motorcycle/moped, bicycle and pedestrian.

deaths were observed in individuals of Danish ethnicity compared to individuals of non-Danish ethnicity. This difference is most likely due to differences in drinking habits such as abstaining from drinking due to religious beliefs.

The main strength of this study is the combination of a variety of independent data sources, including death certificates, autopsy results, media reports and nationwide registers. By reviewing death certificates from all unnatural deaths that occurred during the period of 2010–2019, we minimised the risk of misclassification. Another strength of this study is that we gained access to all death certificates, which enabled complete identification of alcohol-related deaths. We identified more alcohol-related deaths than can be identified in the Register of Causes of Death. Our findings are in line with an earlier study indicating that there is an under-reporting of alcohol-related deaths.¹⁷ Furthermore, this thorough methodology made it possible to characterise alcohol-related deaths among young people in more detail, for instance, describing the circumstances in alcohol-related drownings, than would have been possible using only registry data. To the best of our knowledge, this is one of the first studies to identify alcohol-related

deaths among 15–24-year-olds among all unnatural deaths using nationwide data.

The study has limitations. Most importantly, our count of alcohol-related deaths was conservative for two reasons. First, in defining deaths as alcohol-related, we had to rely on information present in death certificates and autopsies. Alcohol tests are not routinely performed, and in our study, tests results were available in a small fraction of unnatural deaths only (76 of 1067). Of the 76 tests, 75 were positive for alcohol. This makes it highly likely that alcohol could have been involved in other cases too where there was not taken tests. So, we can be quite sure that the true number of alcohol-related deaths was higher than what we found; however, it is harder to know if the underreporting was differential by socioeconomic position. Alcohol misuse is stigmatized and there may be some underreporting especially in individuals of higher socioeconomic position. If so, this would have distorted our finding concerning socioeconomic inequality in alcohol-related deaths in 15–24-year-olds, in a non-conservative direction. Second, our count included deaths caused directly by alcohol only and harm to others were not accounted for. Thus, in cases such as traffic accidents and homicide we counted the deaths of the active

	N (%)		Odds ratio (95% CI)		
	Alcohol-related deaths	Controls	Simple ^a	Adjusted ^b	
Parents' employment status					
Both parents working	54 (46)	756 (64)	1.0 (ref)	1.0 (ref)	
One working one benefit	39 (33)	301 (25)	1.8 (1.2-2.8)	2.0 (1.3-3.1)	
Both benefits	25 (21)	130 (11)	2.7 (1.6-4.5)	4.2 (2.3-7.5)	
Parents' education ^c					
Long	29 (25)	472 (40)	1.0 (ref)	1.0 (ref)	
Medium	64 (54)	583 (50)	1.8 (1.1-2.8)	1.8 (1.2-2.9)	
Short	25 (21)	117 (10)	3.5 (2.0-6.2)	3.9 (2.2-7.0)	
Ethnicity					
Emigrant/parents emigrants	9 (7)	163 (13)	1.0 (ref)	1.0 (ref)	
Danish	114 (93)	1067 (87)	2.1 (1.1-4.1)	4.0 (1.7-9.5)	

Note: 7 individuals had missing information on indicators of socioeconomic position and ethnicity which is why the sum of alcohol-related deaths in Table 3 does not sum up to the total number of alcohol-related deaths in Table 2. ^aAdjusted for matching variables (sex and age). ^bAdjusted for matching variables (sex and age), ethnicity and parents' employment status. ^cThe highest education attained by either parent defined as short (elementary school only), medium (high school and vocational education and short tertiary education) and long (medium and long tertiary education corresponding to university bachelor and master).

Table 3: Odds ratios (95% confidence intervals) for indicators of socioeconomic position and ethnicity, alcohol-related deaths compared to population-matched controls.

Table 3: Odds ratios (95% confidence intervals) for indicators of socioeconomic position and ethnicity, alcohol-related deaths compared to population-matched controls.

parties as alcohol-related deaths and not those affected. For instance, passengers who died in a car accident with a drunk driver was counted as alcohol related; in this case only the driver is included if he or she died in the accident.

A limitation was also that in 60 of 125 alcohol-related deaths, we had information about when the deceased was found and not the exact time of death. Missing data on time of death is common in cases of drownings and other incidences where the deceased is found hours, days or weeks after their death, leading to some uncertainty regarding the time of death.

Implications first and foremost include that reducing binge drinking is an urgent target for preventive strategies and ensuring that such strategies will embrace adolescents and young adults with low socioeconomic background. Binge drinking is frequent in young people in most Western societies and Denmark is taking a lead in this regard: Compared with 29 other European Countries, Denmark has the highest rate of adolescents reporting being intoxicated within the last 30 days. 41% among boys and 40% among girls compared to European average on 14% and 13%.³² Some of the measures that have proven to be effective are increasing the minimum alcohol purchase age,³³ pricing policies,^{34,35} reducing opening hours,³⁶ and responsible beverage service.³⁷ Many of these measures can easily be restricted in Denmark as alcohol can be legally bought from age 16 (as opposed to 18 in most other European

countries) and alcohol is relatively cheap and can be bought 24/7.

From a harms-reduction perspective, the observation that a substantial proportion happened immediately after a party can be used to speculate on initiatives to ensuring that young people get home safely when they have been drinking.

In conclusion, we found that alcohol-related deaths accounted for 12% of all unnatural deaths among adolescents and young adults aged 15–24 years. Males accounted for 84% of all alcohol-related deaths. The most common cause of alcohol-related deaths was transport injuries, followed by poisonings and drownings. Alcohol-related drownings accounted for more than half of all deaths by drowning. There was a clear socioeconomic gradient in alcohol-related deaths and relatively more deaths occurred in adolescents and young adults of Danish as compared to non-Danish ethnicity. There is a great potential to prevent these deaths with prevention strategies targeted at reducing binge drinking and securing that young people get home safely after they have been drinking.

Contributors

JT, UB and SSH contributed to the overall concept of the paper. CHU was responsible for first drafting of the manuscript and literature search. UB and SSH did the manual reviews and JT did the statistical analysis and wrote the final draft. All authors discussed the results and contributed to the final manuscript. All authors have seen and approved of the final text.

Data sharing statement

The datasets generated and analysed during the current study are not publicly available due to sensitivity of the data but are available from the corresponding author on reasonable request.

Declaration of interests

All authors have completed the ICMJE uniform disclosure form at <https://www.icmje.org/disclosure-of-interest/> and declare: no support from any organisation for the submitted work; no financial relationships with any organisations that might have an interest in the submitted work in the previous three years; no other relationships or activities that could appear to have influenced the submitted work.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.j.lanepe.2023.100620>.

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